NATIONAL INSTITUTE OF TECHNOLOGY, PATNA

Mid Sem Jul-Dec 2025

Operating System (CS34111)

Dept of CSE | BTECH - CSE 3rd Sem | Marks 30

Question	CO	BL	Marks
Q1. (a) A semaphore S is initialized to 3. If 4 wait (P) operations are applied on S by 4 different processes, then: a) All 4 processes proceed b) First 3 proceed, 4th blocks c) All block d) Deadlock occurs immediately	2	Un	1
Q1. (b) The Process Control Block (PCB) of a process does not contain: a) Process state b) Program counter c) Page table information d) Source code of the program	1	Rem	1
Q1.(c) The long-term scheduler is responsible for: a) Deciding which process to execute next b) Selecting processes from disk for admission to the ready queue c) Moving processes from ready to waiting state d) Handling I/O device scheduling	1	Rem	1
Q1. (d) Context switching involves: a) Saving the state of old process and loading the state of new process b) Terminating a process and starting another c) Switching from user mode to kernel mode d) Switching between threads of a process	1	Un	1
Q1.(e) Which of the following is true about a thread compared to a process? a) A thread has its own PCB b) Threads within a process share code and data c) Thread creation is costlier than process creation d) Threads do not share the same address space	1	Un	1
Q2. What is the role of process scheduler? Why we need process scheduler in OS? Four processes arrive as follows: PID Arrival (ms) Burst (ms) P1 0 8 P2 1 4 P3 2 9 P4 3 5 Compute waiting time and turnaround time for each process under Round Robin schedule with time quantum = 4 ms	1	Rm+ Ap+	1+1+8
Q3. Define thread and its advantages. What is the disadvantage if multiple process run concurrently? What are the possible outputs if the following two process P1, P2 runs concurrently and they share a global variable x=10. How to prevent circular wait condition to prevent deadlock? Process P1: Read x; x=x/2; Write x Process P2: Read x; x=x+4; Write x	2	Rm+ Un+ Ap	2+2+4+2
Q4. An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes P0, P1, and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution. AVAILABLE X=3, Y=2, Z=3 MAX / ALLOCATION / NEED X Y Z / X Y Z / X Y Z P0 8 4 4 / 0 0 1 / 8 4 3 P1 6 2 0 / 3 2 0 / 6 2 0 P2 3 3 3 / 2 1 1 / 1 3 3 There are 3 units of type X, 2 units of type Y and 3 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state. Which request can be served? Justify your answer. REQ1: P0 requests 0 units of X, 0 units of Y and 3 units of Z REQ2: P1 requests 2 units of X, 0 units of Y and 0 units of Z	2	Un +Ap +Ev	5

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